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Bachelor Thesis

Data Visualization for Augmented Reality Head-Up Displays in Personal Protective Equipment

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1 Introduction

While working in personal protection equipment, it is essential to be able to parse critical information, such as oxygen levels, at a glance. Chemical protective suits in particular reduce peripheral vision and mobility [Murray et al., 2011], making it challenging to operate radio devices and other tools.

A possible solution might be a head-up display (HUD), since it could integrate information into the field of view. Numerous studies confirm an increase in task performance when using a HUD [Kim et al., 2019].

This thesis aims to analyze requirements and current technological issues, as well as propose potential user interface designs for HUDs independent of hardware implementation.

I recommend testing yourself if the design works in the peripheral vision.

2 Chemical Protective Clothing and Its Applications

Scenarios where workers are exposed to hazardous materials require chemical protective clothing (CPC). Protection levels range from limited spray protection (Type 6) to fully encapsulating, gas-tight suits (Type 1) with self-contained breathing apparatus (SCBA)[1] ["DGUV Information 212-019"].

Figure 1: Type 1 chemical protection suit 4005/NEO manufactured by isotemp[2]

Confined spaces are defined as areas in which, through their design or site conditions, "inadequate air replacement rates, or the substances [...] found within them or introduced into them, particular hazards exist or may arise that substantially exceed the normal prevailing hazard potential of workplaces" ["DGUV Regel 113-004"]. Tasks that require entering confined spaces are typically cleaning, inspection and/or maintenance of, for example, tanks in chemical plants, rail cars and highway tank trailers and even sewage containers. Traces of the contents may remain, even if their containers are prepared correctly. These residues might—despite being initially inactive—become dangerous through work procedures (e.g.

[1] A self-contained breathing apparatus consists of a respirator connected to a gas cylinder supplying breathable air. The cylinder is fastened to a harness worn on the back.

[2] Vollschutzanzug 4005/NEO · Chemie · isotemp

welding), agitation or chemical reactions ["DGUV Regel 113-004"], hence requiring the highest levels of protection. Other use cases are emergencies, such as confined space rescue, removing chemical spills or other accidents involving unknown, possibly hazardous substances ["OSHA Technical Manual Section VIII"].

These tasks are highly demanding—both physically and mentally—particularly because of the equipment. Since Type 1 suits are not gas-permeable, they compromise the body's ability to regulate temperature, leading to exhaustion and heat fatigue even faster [Holmér, 2006]. The exertion is exacerbated by heavier equipment and heat stress, ultimately reducing work capacity and performance [Smolander et al., 1984]. The maximum working time in a full-body suit with SCBA is therefore limited and might decrease even more depending on oxygen and exhaustion levels.

Further, the material and shape of the suit limit mobility and the field of view to such an extent that even trained personnel may take twice as long to complete tasks when wearing CPC [Murray et al., 2011].

3 Augmented Reality

Augmented reality (AR) is a subcategory of extended reality (XR), alongside mixed reality (MR) and virtual reality (VR). In AR, virtual objects are "superimposed upon or composited with the real world" [Azuma, 1997], in opposition to virtual reality which—by replacing the real environment entirely—creates the deepest sense of immersion. Mixed reality combines features of both concepts, the main distinction being the ability to interact with virtual objects in the real world [Hari Chandana et al., 2023].

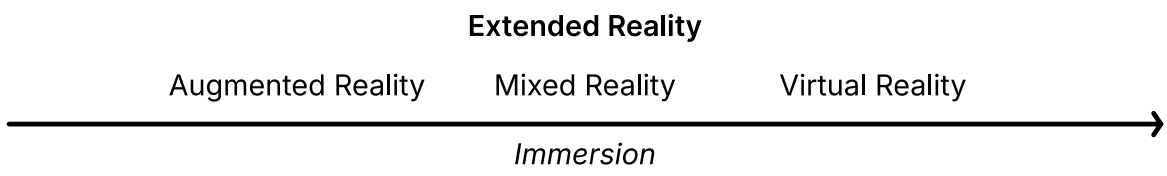

Figure 2: The extended reality spectrum, encompassing augmented, mixed and virtual reality.

Azuma (1997) defines three requirements for applications to be classified as AR. They combine real and virtual content, they are interactive in real-time, and the virtual objects are registered in 3D space. With the technological advancements since the 2000s these requirements are understood more as guidelines rather than rules.

Head-up displays (HUDs) overlay information onto reality in the user's line of sight, allowing them to access necessary information without having to look away from their primary task. Originally used in military aircraft, today HUDs are installed into consumer vehicles as well. In addition to these integrated HUDs, content can be presented on a head-mounted display. In this case, the display is attached to the head, typically as a helmet or glasses, enabling mobile usage. In contrast to the above definition by Azuma (1997), the content displayed on a HUD is two-dimensional and head-stabilized, rather than spatially anchored, i.e., world-stabilized (see Figure 3). Strictly speaking, a HUD therefore does not meet the criteria for AR. However, other research considers head-stabilized presentation of content as suitable for augmented reality [Feiner et al., 1993, Lu et al., 2020].

G. Lin et al. (2021) actually mention a few more advantages:

A head-mounted display can cover one (monocular) or both eyes. Depending on the technology, both eyes receive the same image (biocular) or each will be shown a different image (binocular).

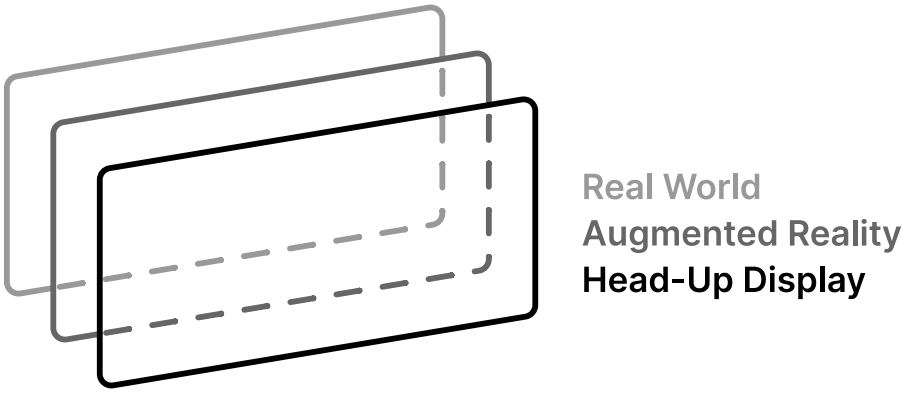

Figure 3: Visualization layers of head-worn displays: reality, Augmented Reality (3D, world-stabilized) and User Interface (2D, head-stabilized).

3.1 Optical See-Through Displays

There are two approaches to combining virtual content with reality. With video see-through displays, information is superimposed onto a video stream captured by a camera mounted on the device. These headsets fully enclose the head to block external light. In contrast, an optical see-through (OST) system allows the user to view reality directly. Here, the overlay is created by optical combiners blending light from the image source and the environment together.

Since video see-through displays completely block the view of the outside world, it is impossible to see anything in the event of a failure. Given the intended use cases, this represents a major safety issue, especially since the headset cannot be removed while wearing a fully encapsulating suit.

Syberfeldt et al. (2017) recommends a field of view of at least 30° for industrial applications. Microsoft's HoloLens 2 features a 43° by 29° field of view, while a few other systems offer around 40°.[3]

As with all head-mounted devices, walking causes them to shake, which affects legibility [Janaka, Zhao, and Chan, 2023]. Especially thin, horizontal lines tend to disappear optically due to the up- and down motion.

Apart from these technical limitations, see-through displays present a number of major problems affecting interface design.

[3]Compare all AR Headsets & Glasses - VRcompare

3.1.1 Ambient Light and Contrast Issues

Regardless of the optic technology used, images are created by combining light, which means the display has to compete with ambient light for visibility. This results in a significant loss of contrast, especially outdoors. Kerr et al. (2011), for example, report that participants repeatedly used their hands to shield the display from external light to be able to read its contents.

Many manufacturers of optical see-through displays equip them with shaded lenses to improve the contrast somewhat [Itoh et al., 2021]. Unfortunately, current OST technology is incapable of producing the contrast levels required for outdoor use [Erickson et al., 2020]. Technical solutions currently in development include OSTs that can partially occlude areas of the display to block light. However, these are still very experimental [Kiyokawa et al., 2000, Davari et al., 2020].

As a hardware-independent solution, Tanaka et al., 2008 propose a method for automatically placing content where readability is better, i.e. darker segments of the background. Such concepts might not be suitable for safety-critical scenarios, as users rely on spatially anchored elements to quickly acquire information.

3.1.2 Color Blending

In the additive color model, the combination of all wavelengths of light produce white, while black is created by the absence of light. When physical screens display black, the corresponding pixels do not emit light, so the actual color of the display material is visible. Since OHMDs do not have a physical screen, "black" is transparent [Gabbard et al., 2013]. This also means that darker colors are not available for design on OHMDs, as they are mostly translucent. Additionally, the level of transparency affects the extent to which the background distorts the color [Itoh et al., 2021].

Gabbard et al. (2013) additionally found that, depending on the background, colors may shift in hue or washout due to their luminance value and/or chromaticity. Both of these effects result in colors potentially being misinterpretated or not being visible at all.

3.2 Methods of Interaction

Due to not being constrained to two dimensions, XR applications can explore new types of interaction, such as gesture recognition, speech recognition and gaze tracking. However, industrial use cases require hands-free interaction methods and the high noise levels expected in these environments limit the possible forms of interaction to eye tracking. Even with these constraints, Hari Chandana et al. (2023) emphasize the importance of interactions being intuitive and feeling natural.

Gaze-based interaction is enabled by eye tracking technology. In addition to tracking eye position, it analyzes where the user is looking and how long.

Eye tracking could be used to dim the HUD while the user does not look at it. This would decrease distractability, and reduce glare in darker environments.

4 The Human Visual System

4.1 The Field of View and Peripheral Vision

The field of view spans approximately 200° to 220° horizontally and around 125° vertically, with the vertical field of view extending slightly further downward [Loschky et al., 2017]. The central vision covers about 5° around the point of fixation, where acuity is highest [Rosenholtz, 2016, Loschky et al., 2017]. The remaining field is the periphery, with boundaries at 30° and 60°, dividing it into near, mid and far peripheral vision.

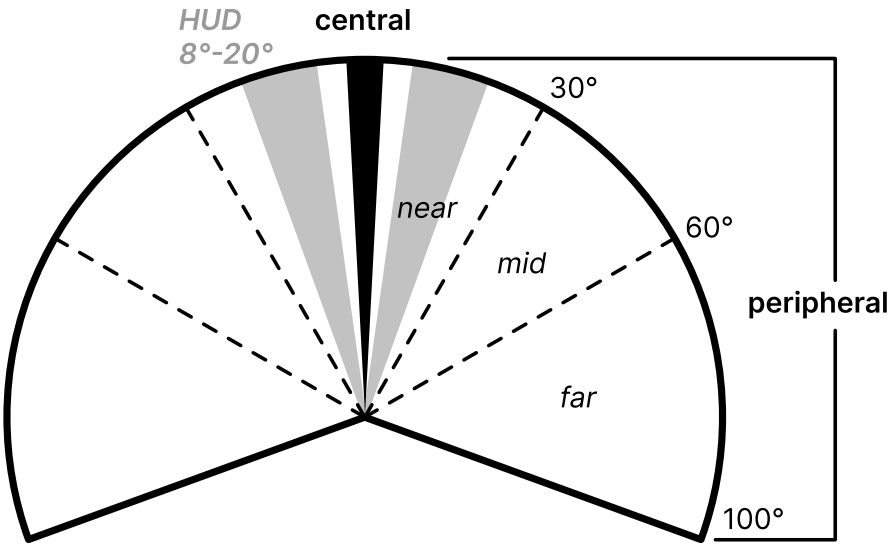

Figure 4: Simplified illustration of the horizontal field of view.

Despite visual acuity and other functions deteriorating in the periphery, the area in front of us continues to appear in high resolution, with high contrast and color saturation [Loschky et al., 2017]. According to Rosenholtz (2016), the decline in performance of the peripheral vision is smaller than commonly assumed. A scene does not require many details to still allow object recognition and orientation, which is why image thumbnails remain recognizable despite their small size.

In addition, peripheral vision is particularly well suited for detecting motion, provides better vision in low-light conditions [Johnson, 2014] and contributes to functions like orientation, navigation, balance, and hand-eye coordination. A restriction of the peripheral visual field severely impairs the performance of these functions, the effect increasing with the amount of vision obstructed [Alfano and Michel, 1990].

Color sensitivity decreases toward the periphery, red-green moreso than blue-yellow colors and sensitivity to luminance. Hansen et al. (2009) found that color perception is possible even at an eccentricity of 50°, provided the stimulus is sufficiently large. Most modern OST-HMDs feature a 40° field of view (see Section 3.1), meaning the entire available color spectrum can be utilized.

Rosenholtz (2016) suggests that crowding is the biggest issue in parsing information located in the periphery. When visual elements are placed too close together, they blend into a single entity, making it harder to distinguish them from one another. This effect is more pronounced if the target and its neighboring elements are similar (e.g., same color and shape) [Pelli and Tillman, 2008]. The farther elements are from the center of vision, the greater the distance between them must be.

4.2 Visual Attention

The retina transmits too much information to be processed all at once. This bottle neck is resolved by visual attention, which focuses resources on just a fragment of the current visual stimuli [Anderson et al., 2005, Rosenholtz, 2016].

Two processes determine where this attention is focused next. Bottom-up attention occurs involuntarily, triggered by external stimuli. In the top-down process, the gaze is directed toward objects related to current needs or goals. Interestingly, top-down attention reduces the impact of distracting stimuli while consciously focusing on a visual target [Connor et al., 2004].

An object is referred to as visually salient, if it attracts attention by being distinct from its surroundings in some way, without specifying which characteristics cause it to do so.

When multiple stimuli compete for attention, even salient targets might be ignored if they are irrelevant to the current goal. This is referred to as inattentional blindness [Jensen et al., 2011].

5 Attention Management

After discussing the mechanisms of perception and attention, the question arises as to how they can be utilized and which aspects need to be taken into consideration in order to do so.

5.1 Situation Awareness

Making good judgements becomes increasingly difficult when performing demanding procedures while trying to maintain awareness of the surrounding area. The risks associated with these activities prohibit human error, further increasing stress levels. Endsley (1995) highlights the role of situation awareness as a foundation for good decision-making in dynamic situations. She outlines three levels of SA, namely perception, comprehension and projection, of which perception is the most significant for UI design. This stage requires collecting the information relevant to the situation. In level 2 this information will be analyzed to form "a holistic picture of the environment, comprehending the significance of objects and events." Finally, likely outcomes can be determined by understanding the dynamics of the situation.

It is evident that situation awareness is therefore related to the processes of attention described in the previous section. Subsequently, this illustrates the necessity of a thorough understanding of the pieces required for achieving good situation awareness to design a well-functioning HUD. The amount of information that can be rapidly and accurately obtained is directly influenced by how it is presented. In addition to prioritizing and organizing that information around task objectives, Endsley (1995) mentions the advantage of systems providing processed information rather than the data it is based on (e.g. displaying remaining time based on current oxygen amount and usage).

She emphasizes that special attention should be paid to reserve highly salient features to critical events that truly necessitate changing the course of action. Further, it is vital to maintain a balance between showing all relevant information, while simultaneously avoiding visual overload. While looking for inaccessible information, other important content may be overlooked. Conversely, "filtering of extraneous information (not related to SA needs) and reduction of data (by processing and integrating low-level data to arrive at SA requirements)" is necessary to prevent clutter. This reiterates the importance of accurately evaluating and prioritizing

task-relevant information.

As Endsley (1995) points out, considering these recommendations supports assessment of the interaction of all components as a whole, since emphasizing some elements inevitably comes at the expense of others. Evaluating the design concept by the amount of situation awareness it provides serves as the last step of ensuring the effectiveness of the system.

5.2 Peripheral Displays and Glanceability

The requirements of situation awareness make clear that information has to be quickly accessible, without straining cognitive capacity or being overly distracting. Peripheral displays do just that—keeping relevant information accessible while offering the choice of whether or not to continue with the current task [Matthews et al., 2005]. On the other hand, so-called notification displays actively interrupt the user to prompt necessary action.

Building onto this categorization, Matthews et al. (2006) introduces the concept of glanceability—a term used to describe information that is quickly and easily comprehended once it is paid attention to. Glanceability is directly impacted by the displayed amount of items, their salience, etc.—i.e. the attentional limits described in Section 4. In this context, Matthews et al. (2006) emphasizes that visual components need to "balance distinctiveness with too much demand for attention."

Lu et al. (2020) note that to design effective glanceable displays, it is necessary to consider both the visual presentation of information and how it is accessed. They present two novel concepts for hands-free interaction that implement the principles of glanceability and compare their performance to a regular HUD. Participants in the study were asked to follow a virtual mannequin at a specified distance for the primary task, while being asked questions about the information shown on the display.

The eye-glance interface corresponds to a conventional HUD, in which elements are fixed at the edges of the display's field of view. Although continuously displaying the information makes it easily accessible, it can lead to information overload or otherwise be distracting. Additionally, restricting the peripheral field of view can impair perception in that area.

The visual components of the head-glance interface are placed just outside the

display's field of view and are accessed by turning the head. Without this movement, the elements are not visible at all. This approach requires an additional sensor to track the head movement relative to the position of the body. According to Lu et al. (2020), the interface caused frustration among the participants. However, they theorize that this was due to the frequency of the questions asked, which led to numerous head movements within a short amount of time. In addition, participants found it mentally taxing to remember the location of specific information.

In the gaze-summon interface, visual targets are placed near the position where the interface elements would be located. If these points are fixated for a certain amount of time, the information appears at that location. However, in industrial environments this delay may not be justifiable in the intended applications. Additionally, Lu et al. (2020) report that keeping the gaze on target might lead to eye strain if the display is used frequently. There was also an increase of false positives when participants took a turn, caused by reflexively looking into the intended direction, combined with eye-tracking jitter from the rapid head movement.

These methods offer an interactive approach to countering information overload. That said, visual elements must also be designed, and in such a way as to draw the user's attention to the relevant information.

5.3 Interruption

Matthews et al., 2006 "Design variables used for capturing user attention include abrupt onset, flashing, bouncing, and other motion."

Notification levels equal importance of information, corresponding to action priority

Matthews et al. (2004) present a "Peripheral Displays Toolkit", which helps to assign notification levels to information, shows how these transitions between those levels should look, and how information has to be abstracted to be "glanceable". "Abstraction involves extracting features or reducing the fidelity of information so that it is easier to read "at a glance" than the raw input. Abstraction enables lower-attention consumption of information." Notification levels correspond to information priority. distinction again between getting attention and just monitoring They categorize attention into inattention, dividided attention and focused attention, where the two latter kinds are how humans consciously perceive. The notification levels presented are ignore, change blind, make aware, interrupt, and demand action.

Demand action even requires the user to do something to stop the alarm. "While it may seem contradictory for a peripheral display to attract focused attention, this enables a display to be peripheral when it is not displaying important information. A watch alarm only requires focused attention when it goes off, alerting the user that it is time to go somewhere. If a user could not count on the alarm to attract her full attention, she would pay more attention to the watch." With transitions, effects that capture an appropiate amount of attention can be created. Attracting focal attention equals interrupting the user, usually with abrupt transitions. When aiming for divided attention, abruptly update smaller areas of the screen or use slow movements.

Notifications are naturally associated with the cost of potentially distracting or completely interrupting the user. Janaka, Zhao, and Chan (2023) suggest using animation to soften the impact a little. Fade-in animations reduce interference, though they are dependent on duration, location, and primary task complexity. The longer the fade-in, the less noticable the notification becomes. Additionally, the duration is affected by external lighting. "Although fade-in animation can reduce the interference to primary tasks, its practical usage should depend on the utility of notifications [21, 40]. As expressed by two participants, how notifications appear can signal their level of importance and urgency; slow fade-in notifcations can and are suited to signal lower levels of urgency [18]. Although perceived urgency and importance are infuenced by fade-duration, the maximum delay of notifications was 4-seconds. This may mean that fade-duration is not the biggest fact in attendance to notifcations."

6 Data Visualization

Visualizating data transforms data into information. Data itself is just noise, whereas information holds value. This value is created by the intended context, which determines in what way the data needs to be transformed [Tergan and Keller, 2005].

6.1 Guiding Visual Search

Information needs to be located before it can be interpreted. To make this happen as quickly as possible, certain design principles can be utilized.

To this end, Shao et al. (2023) present the 3V method, which refers to visual hierarchy, weight, and flow respectively. Visual hierarchy on the interface level uses unity, contrast, and order to sort and group information. The visual weight is described by the features of and space occupied by elements. It works through the distinction of objects, achieved by differing presentation. More visual weight means the object attracts more attention. The visual connections between elements describe the visual flow. If implemented correctly, flow guides the gaze to target information.

They evaluated this framework by testing visual search performance of four different label arrangements. The labels consisted of a string of numbers, and described boxes chaotically arranged on the screen. They found that visual search is more efficient, if the labels are arranged in vertical columns on the sides of the screen. This facilitates scanning of the content. Two reasons: if the labels are read left to right, it is easer to distinguish the start and end of them. Although the eye moves faster horizontally than vertically, which makes it easier to control jumps between rows.

In a similar vein, Matthews et al. (2006) explain: "Visuals that support top-down search will be easy to cognitively interpret. Visuals that support bottom-up search enable relevant information to pop-out."

In summary, even when a user is consciously (i.e., top-down) searching for a specific element in the user interface, a well-structured design can leverage bottom-up attention to guide this visual search [Shao et al., 2023].

6.2 Facilitating Rapid Comprehension

To make information glanceable, concepts of data visualization can be leveraged.

Wilke (2019) lists position, shape, size and color as basic elements for visualization. Position describes the placement of data, related to other points or axes of a diagram. As for color, hue and value should be distinguished, since they are suited for different kinds of data. Hues tend to work better for distinctive categories. Value for continuous data GRADIENTS, the eye is much more sensitive to differences in lightness (see Section 4). Hues can also be used for gradients (e.g. rainbow), but require much more intention to work.

Each of these elements can be used to encode information. Which element is used for which information influences the type of visualization. Importantly, this mapping must be unique, else it becomes ambiguous and obscures the meaning [Wilke, 2019].

Information can be abstracted when the represented concept is simpler (e.g. ambient motion detector), in which case primitive elements of design, such as color and movement can be sufficient.

6.3 Visualizations for Single Value Data

Aside from design fundamentals, most specialized resources discuss visualization types intended for interrelated data. Interpreting such complex information requires time and cognitive capacity unavailable in scenarios such as those mentioned in Section 2. In these situations, a clear picture of the current status must be available at all times, which only allows for individual data values and therefore limits the number of possible visualization types.

Ferdio, an agency specialized on infographics and data visualization, has created a library of visualization types[4]. In this library, out of a 166 options, only nine are for single data (see Figure 5).

Icons and Scaled-up Numbers. highlight a single value Often used when the number does not need a context or a comparison icons serve as description or label

[4]Data Viz Project | Collection of data visualizations to get inspired and find the right type

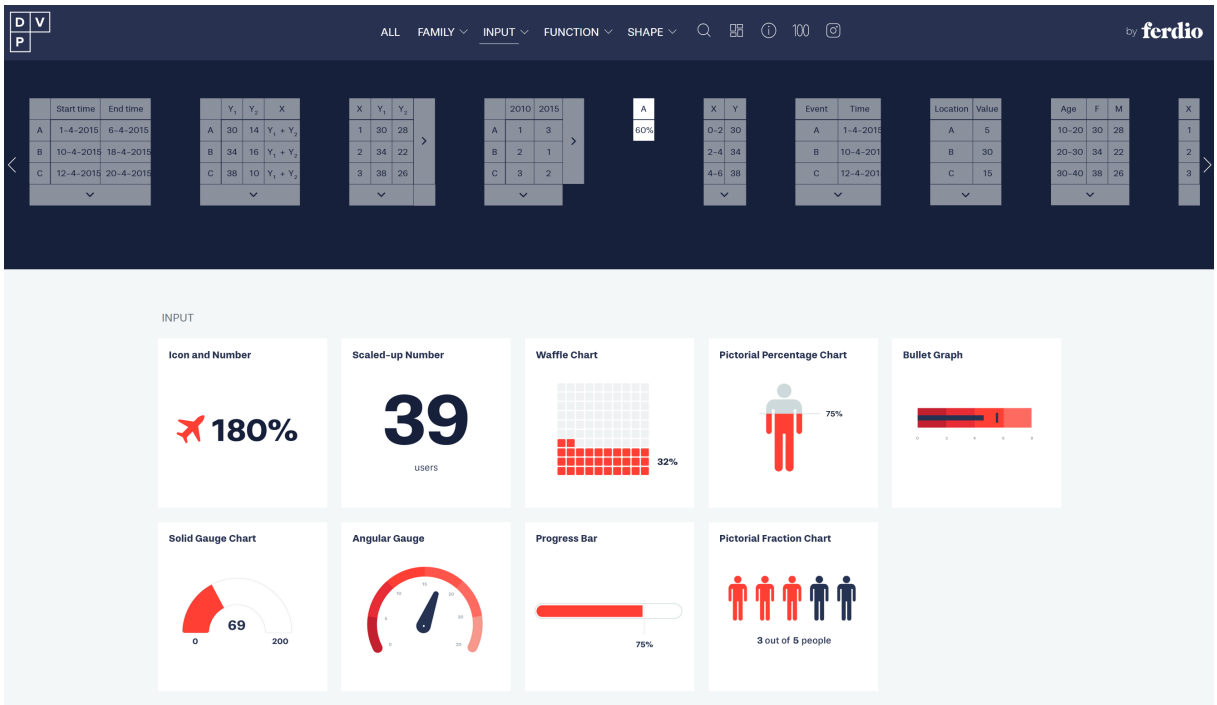

Figure 5: The options for single data (highlighted in the menu bar) in Ferdio's data visualization library.

Differential Indicators. These are a type of status indicator, and meant to visualize changes of information. arrows stock values, red green

Gauges. solid necessarily show specific data/numbers, more for abstract status evaluation Angular Gauge uses a radial scale for specific data points (e.g. temperature) dial over a radial scale with defined limits

According to Xia et al. (2025), bar charts outperform radial visualization.

Status Bar. A progress bar is a graphical control element used to visualize the progression of an extended operation, such as a download, file transfer, or installation. Sometimes, the graphic is accompanied by a textual representation of the progress in a percent or quantitative format.

7 Design Guidelines

To facilitate situation awareness, the HUD should show as much information as possible without causing fatigue or sensory overload. The displayed information should be comprehensible in as short a time as possible. In an emergency, the system must be able to get the user's attention, without hindering decision-making or interrupting the primary task completely.

7.1 Color and Contrast

As mentioned in Section 3, the use of colors in HMDs intended for daylight use is primarily limited by current optical see-through technology. Gabbard et al. (2022) recommend to prefer colors with high source luminance, as these are less prone to color shifts. Furthermore, if a color typically represents a specific meaning and cannot be guaranteed to be perceived correctly, alternative colors that can meaningfully convey that message should be considered. Red is specifically cited as an example, as it is highly susceptible to color shifts due to its low luminance.

The library chroma.js[5] has a funtion to sort colors by luminance in the RGB color space, based on the Web Content Accessibility Guidelines 2.0.[6] Sorting the base colors red (#FF0000), green (#00FF00), blue (#0000FF) and their intermediaries cyan (#00FFFF), magenta (#FF00FF), and yellow (#FFFF00) puts them into the following order (brightest to darkest): yellow, cyan, green, magenta, red, and blue.

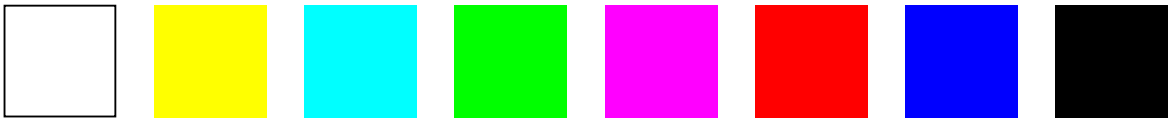

Figure 6: Base colors sorted by luminance.

Limiting the UI to a single color can ensure that the design functions effectively in every respect (e.g., by using motion to draw attention rather than color). Not encoding information through colors also avoids potential semantic misinterpretations that could be caused by the previously mentioned color issues. In addition, monochrome displays are more affordable and therefore offer economic advantages, since replacement costs should be kept as low as possible in cases where equipment may be damaged or contaminated.

[5] chroma.js api docs!
[6] Web Content Accessibility Guidelines (WCAG) 2.0

7.2 Typography

Aside from screen shake, reading performance is impacted by lighting and background texture, reiterating the importance of proper contrast.

To find requirements for font size, Renkewitz et al. (2008) tested to what extent the background and font colors affect text recognition. They set a benchmark of two seconds as threshold for fast recognition, which resulted in recommended font sizes from 23 to 42 pt, depending on color and background. However, these numbers may have to be adjusted for an OHMD, since they used a video see-through system, and red and blue as colors, both colors with lower luminance.

As mentioned in Section 3, thinner lines tend to disappear due to screen shake. Matsuura et al. (2019) confirm that heavier font weights improve readability. Naturally, the same requirements exist for stroke width.

As example, the typeface "Inter" fits the requirements well. It has a uniform thickness of strokes, a square letter form and good spacing between its lines.

7.3 Iconography

Janaka, Zhao, and Sapkota (2023) tested if, and under which conditions icons perform better than text. They found that icon-augmented notifications enabled significantly higher primary task performance than text only and reduced the level of interruption while maintaining reaction and comprehension. They also found icons are more comprehensible when task complexity increased or demanded more attention. Additionally, the notification were "easier to perceive in suboptimal conditions", including outdoors when walking.

Naturally, this requires the icons to be well-designed. To that end, H. Lin et al. (2016) tested versions of icons against each other. The differences tested were polarity, plane- and shape-based, and the presence or absence of borders. Since line-based icons dissect the space more frequently, they increase visual complexity, which impairs recognition performance. Overall, white icons on black background performed best.

Another reason to prefer shape-based icons is the requirement of thicker line widths which might impact an icon's readability. Simpler icons with varied silhouettes offset this.

Figure 7: Line-based (left) and plane-based (right) icons.

7.4 Layout and Spacing

The ability of (binocular) head-mounted displays to display content anywhere in the field of view always entails a trade-off between immediate information access and an unobstructed central line of sight. In use cases involving a great deal of physical activity, a clear field of view should of course be given greater priority.

Chua et al. (2016) investigated the positioning of content on a monocular OST-HMD worn in front of the right eye. The display was divided into a 3×3 grid, spanning 12.5° vertically and 17° horizontally in both directions from the center. The top and most peripheral positions were found to be more comfortable and unobtrusive, although middle and bottom center positions were noticed quicker. The center-right is recommended as best balance between both usability and comfort. They mention that the ideal display position is highly dependent on task requirements (e.g., center is better when driving, secondary information better in peripheral).

G. Lin et al. (2021) divide HMD applications into four task categories: single (continuously focusing on display), alternating (switching back and forth between display and reality), background (passively monitoring the display for notifications) and dual (displayed information supplements physical primary task, e.g., captions). Similar to Chua et al. (2016), they mention that optimal positioning depends on task type. Participants preferred 5.4° to 14.6° horizontally, farther out was uncomfortable.

Ultimately, "there may be an ideal, general purpose HWD position between 8 to 20° temporally offset from line-of-sight. Horizontal FOV would thus be 12°."

Dowell et al. (2002) mention 8° around the line of sight should be kept free, since it "can induce cognitive tunneling and impair performance on the task over which the HUD is superimposed."

Crowding (see Section 4) affects perception the most. It seems advisable to

prefer wider spacing between elements.

Additionally, the number of components per side should be limited to three at most, anchoring them spatially to the corners and sides of the display and keeping the center of vision free. **<empty citation>**

7.5 Notifications

Critically examine how information should be handled. Does the user need to be interrupted and does he have to do sth about it?

Dont overdo it with attention-grabbing features, the more contrast to other elements, the more they pop. If everything's the same risk inattentional blindness

The billboard style occludes much of reality around the text, which should generally be avoided. However, exactly this feature makes it suitable for high priority notifications.

Information overload

8 Conceptualization

For printing, the color of the designs was inverted to black on white. A UI example in original colors is included as Figure **??**

8.1 Relevant Parameters

A prioritization of elements is necessary for decisions on visual hierarchy, as well as which ones can be left out if there is not enough space.

	Display Mode	Visualization
Oxygen	Continuous	Status Bar, Value
Hazard Proximity	Pop-Up	Icon
Warning Notification	Pop-Up	Billboard
Working Time	Pop-Up, Continuous	Countdown
Heart Rate	Continuous	Icon, Value, Trend
Ambient Temperature	Continuous	Icon, Value, Trend
Humidity	Continuous	Icon, Value
Checklist	Continuous	Line Highlight
Tool Help	Pop-Up	Icon, Value

Table 1: Parameters chosen for visualization, ordered from most to least important.

Continuous means that the component will be displayed permanently. Parameters marked with pop-up will be displayed occasionally when relevant.

8.2 Components

8.2.1 Visualizing Oxygen Usage Through a Status Bar

Since the oxygen value is the crucial factor whether or not to continue work, this is the component taking the spotlight. As the most critical parameter, this component will be visible in any version of the HUD. To check the current status as quickly as possible, the value will be visualized with a status bar.

Since mechanical gauges with a pointer show visual progress by default, a radial version of the component was included. However, as mentioned in Section 6, linear design enable much quicker parsing of the information. Compared to the linear version, it also takes up much more space to make the information readable.

Figure 8: The initial concepts for the status bar, considering overall shape.

The second version places the value below the bar, on the right. When the amount of oxygen decreases, the empty portions of the bar are only a contour. Thus, the contrast between value and bar becomes stronger. The label is placed in front of the bar so they blend together, making it less salient. Additionally, the status bar was divided into quarter segments, enabling more accurate estimation of the amount left, without looking at the value.

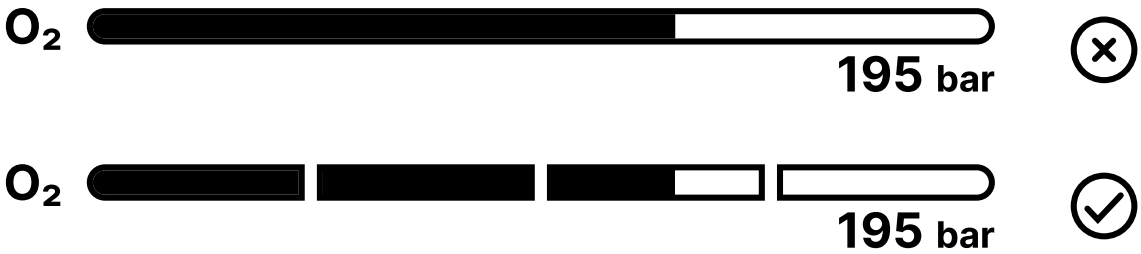

Figure 9: The modified status bar and the following segmentation.

Depending on available screen space and layout configuration, a vertical version of the component might be needed in addition to the horizontal one.

Grouping the label and value on one side saves space (left and middle of Figure 10). Aligning them at the top and bottom respectively accounts for visual crowding, making the value more distinctive. The oxygen value is located so that the status bar is seen first, i.e. if the component is placed on the left side, the bar will be on the right to maintain an optical guide.

8.2.2 The Base Component

This element is built with a simple configuration for secondary information that is helpful, but does not need to be prioritized. This includes data like heart rate, but also environmental information such as ambient temperature or humidity.

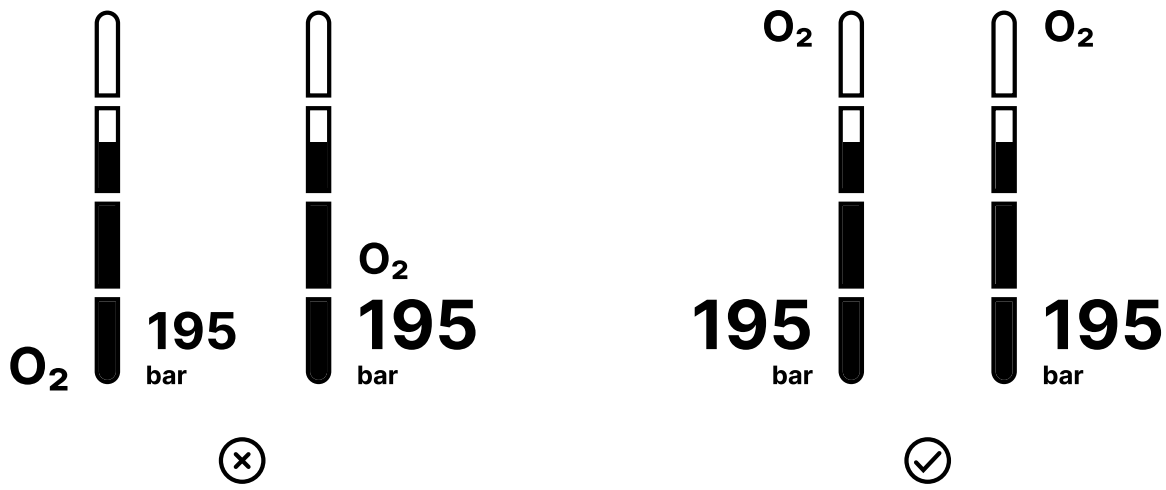

Figure 10: Development of the vertical status bar.

Values that are not to be visualized are similar enough to be based on a universal base component. Apart from the actual parameter value, any of these elements can be hidden. This allows flexible configuration depending on the type of information.

Figure 11: The base component with configurable icon, parameter value, unit and trend indicator.

The icons replace labels, and should therefore be as unambiguous as possible. Leaving the unit out saves space, though in some cases, it supports the identification of the value (e.g., degrees Fahrenheit or Celsius).

The trend indicators are displayed based on the difference of the current and previous value. The value increasing or decreasing an unusual amount might indicate events to take notice of. For dynamic elements like these, the frequency of changes should not be too high to avoid distraction. Comparing the short and long arrows, the long arrows are better readable when out of focus.

All of these values are displayed continuously, with the exception of the time component. It will only be displayed after a configured amount of time. The element popping up is meant to be perceivable in the peripheral vision, marking (for example) the halfway point, encoding this action with information. Since working

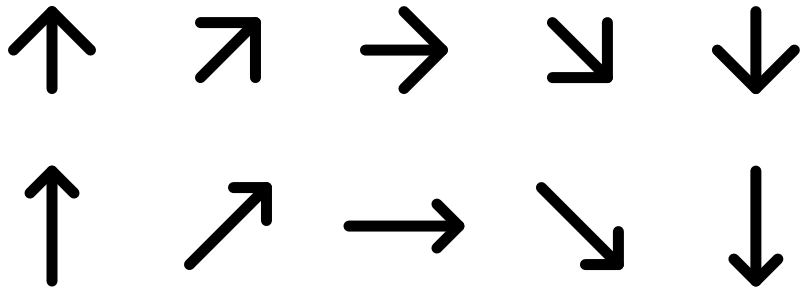

Figure 12: Comparison of short and long arrows.

time is limited by various parameters and therefore not fixed, the component is formatted as a countdown. To avoid pulling unnecessary attention, the display is limited to minutes, rather than ticking down by the second.

8.2.3 Notifications and Pop-Ups

Referencing typical industrial warning signs, the hazard icons were created in a triangular and a diamond-shaped version. Comparing them however, reveals that for the same size, the diamond shape appears smaller. This also means that there is less space for the graphic inside it. Avoiding unique silhouettes unites them in their function. For peripheral distinction, their placement is therefore more important. The icons are placed close together so that they are visible at the same angle if multiple hazards are present. Similar to a car's dashboard warning lights, each of them has a fixed position, facilitating rapid information acquisition. With regular notifications—depending on which one appears first—the order of content would be different.

The core example of the notification system is the high priority warning. This component is designed with a so-called billboard style, meaning text with a surrounding box. The big, bright area ensures high visibility even if the pop-up is not in the center of vision. Since it is the highest priority, it is the only element placed in the upper central screen area. To avoid blocking central vision for too long, the pop-up will transform into a smaller version located at [...].

Figure 13: Hazard indicator icons.

8.2.4 The Checklist Component

Since this element is text-based, it relies less on peripheral vision. To anchor the gaze to the current task, it is centered vertically and more distinctive than the other displayed tasks.

Note that this component is meant as a visual example, which is why certain aspects of functionality might not work in regard to handling the list. (e.g. if only two next tasks are displayed, there is no way to see if new tasks got added → task number display in dynamic situations, generally better for protocol).

The icon to the right is an anchor for possible gaze-based interaction, in this case to show more details. This design keeps the current task centered in the detailed view as well for faster orientation. This also mitigates side effects from possible false positives—i.e. if the system mistakenly opens the detailed view.

8.3 UI Examples

Note: the following illustrations are not to scale.

The maximum variant of the interface was designed with full functionality in mind,

based on Microsoft's HoloLens 2. It features a horizontal field of view of 43°, a resolution of 1440 by 936 pixels per eye, and provides full color availability (within the technical limits mentioned in Section 3).

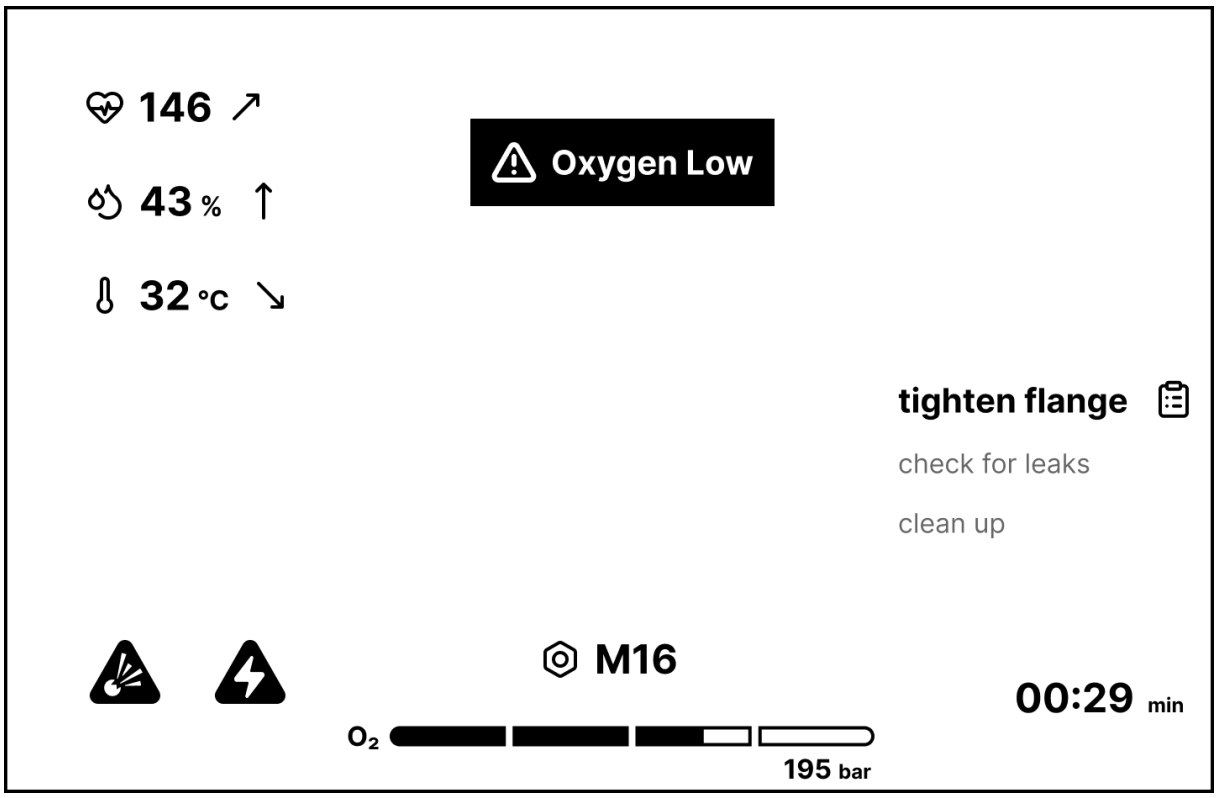

Figure 14: Full-featured UI concept for a binocular HUD.

While the oxygen component is the most important, it is not necessarily the one that is checked most often; however, it should be the easiest to check. Therefore, it is placed in the lower center. Occasional information pop-ups are placed above the oxygen bar, for the same reason: looking down is faster than looking up.

The minimum version is based on the Google Glass, featuring a monocular display with a resolution of 640 by 360 pixels.

One could argue that some overlap of the components would be useful to make better use of the limited space. However, this would disrupt the grid pattern and the visual guides, resulting in longer search times.

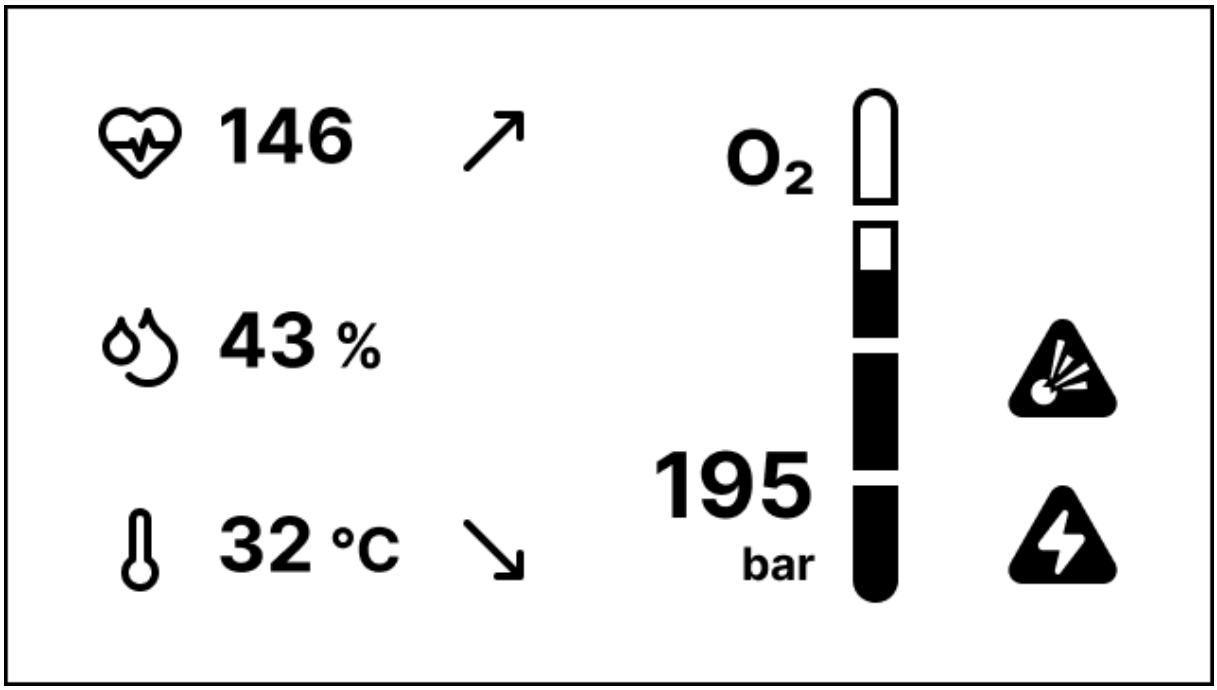

Figure 15: UI concept for a monocular HUD, worn in the upper left of the field of vision.

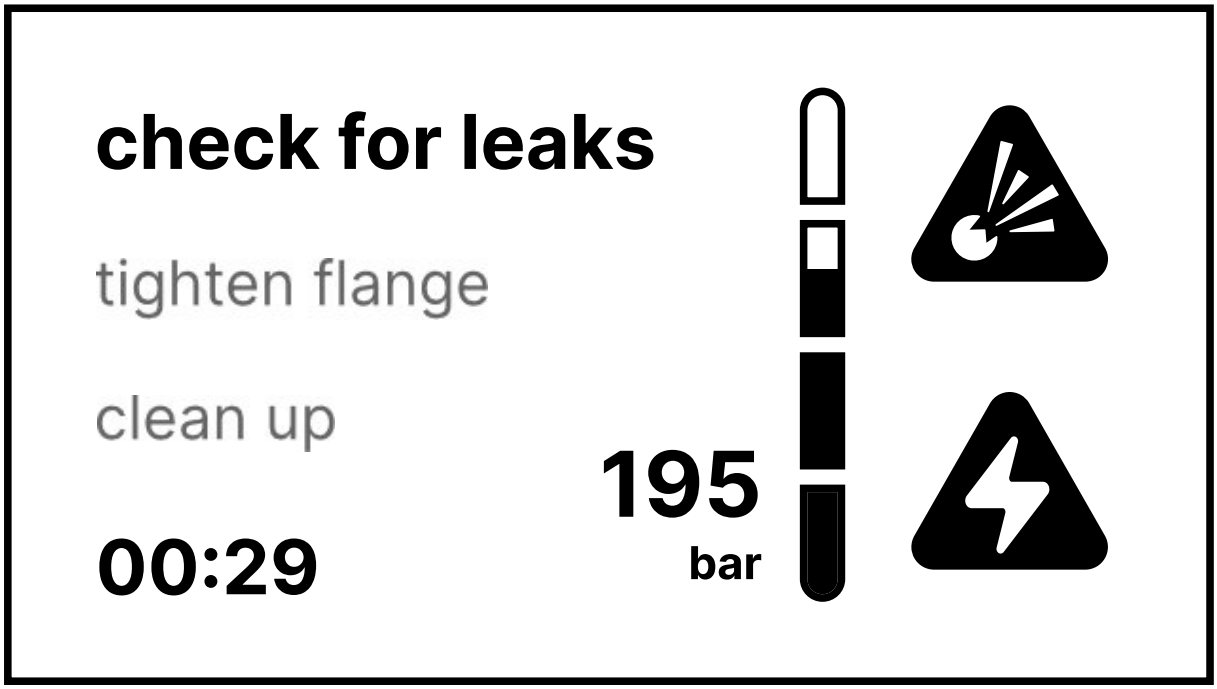

Figure 16: Overlap of the checklist and oxygen bar.

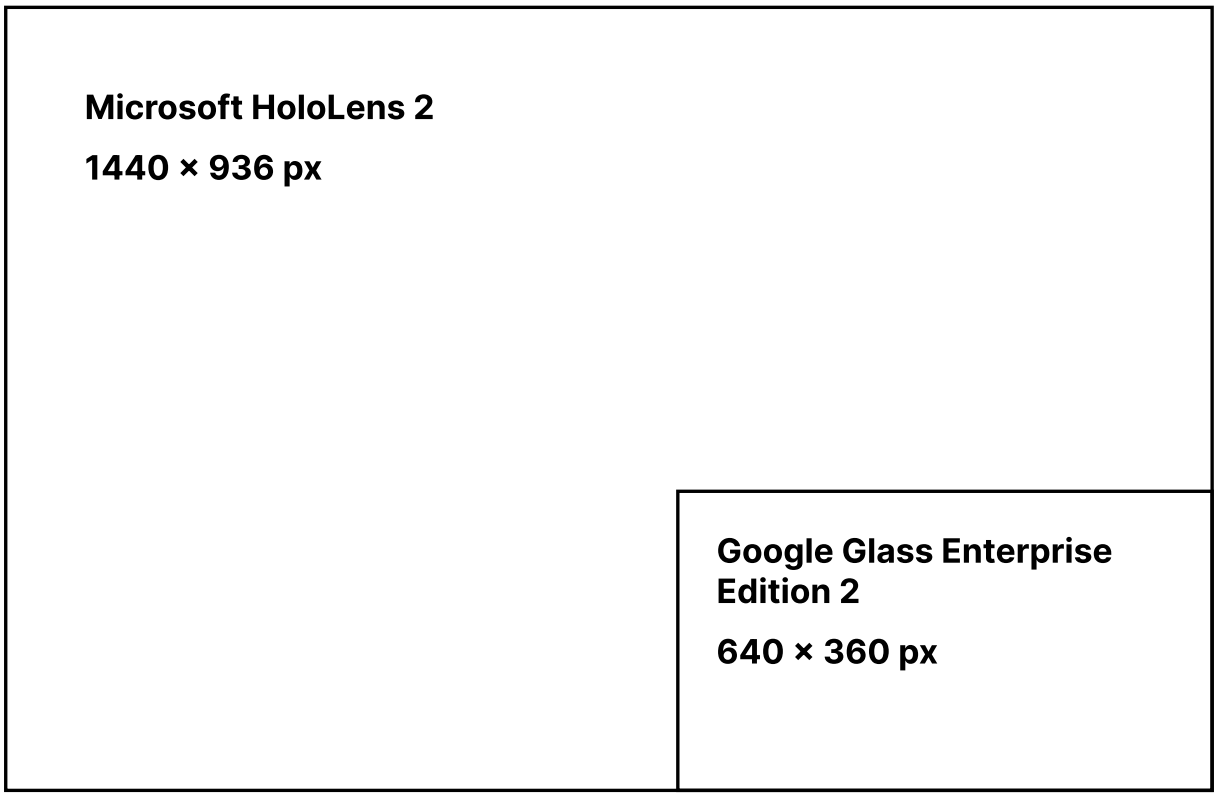

Figure 17: Size comparison of the Microsoft HoloLens 2 and the Google Glass Enterprise Edition 2.

9 Conclusion

Outlook

animation complete hands-free system attaching to control system multimodality: ZHU 2025

testing and iteration: Hari Chandana et al., 2023

Appendix

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